



Making a journey

Task A: Calculating journey times for different transport

1) The table shows the distances of three **commutes**.

Fill in the blanks in the table by calculating the estimated journey times using each method of transport.

method of transport	average speed (km/h)	estimated journey times (to the nearest minute)		
		2 km journey	8 km journey	16 km journey
walk	5	24	96	192
cycle	20	6	24	48
drive	35	3	14	27

Which method of transport would you choose to use for each length of journey?

Answers may differ depending on personal preferences.

Examples:

"I would prefer to always use the fastest mode of transport. So I would drive for each."

*"I like to walk. But walking for over an hour could be impractical for **commuting**. So, I would walk for 2 km journeys and cycle for 8 km or 16 km journeys."*

*"I would like to spend a little bit of time **commuting**, but not too long. An ideal amount of time for me would be between 15 and 30 minutes. So, I would walk for a 2 km journey, cycle for an 8 km journey and drive for a 16 km journey."*



Task B: Calculating carbon emissions for journeys

The amount of CO₂ emissions per mile can be estimated by using the formula:

$$\text{emissions per mile (grams)} = \frac{\text{CO}_2 \text{ per gallon of fuel (grams)}}{\text{fuel efficiency of the vehicle (mpg)}}$$

- 1) A person lives 6 miles from their place of work and travels by car.

The average CO₂ emissions for a petrol car is 9 kg per gallon.

On average, their car travels 45 miles per gallon.

They work 223 days per year.

Calculate the amount of CO₂ emissions produced per year from **commuting** by car. Give your answer in kilograms and accurate to 3 significant figures.

Emissions per mile: $9000 \div 45 = 200 \text{ grams of CO}_2 \text{ per mile}$

Emissions per journey: $200 \times 6 = 1200 \text{ grams of CO}_2 \text{ per journey (1.2 kg)}$

Emissions per day: $1.2 \times 2 = 2.4 \text{ kg of CO}_2 \text{ per day}$

Emissions per year: $2.4 \times 223 = 535.2 \text{ kg of CO}_2 \text{ per year}$
 $= 535 \text{ kg of CO}_2 \text{ per year (to 3 s.f.)}$

- 2) 40 people live 6 miles from their place of work and travel on the same bus.

The average CO₂ emissions for a diesel bus is 10 kg per gallon.

On average, the bus travels 8 miles per gallon.

They work 223 days per year.

Calculate the amount of CO₂ emissions produced per passenger per year from **commuting** by bus. Give your answer in kilograms and accurate to 3 significant figures.

Emissions per mile: $10\,000 \div 8 = 1250 \text{ grams of CO}_2 \text{ per mile (1.25 kg)}$

Emissions per journey: $1.25 \times 6 = 7.5 \text{ kg of CO}_2 \text{ per journey}$

Emissions per day: $7.5 \times 2 = 15 \text{ kg of CO}_2 \text{ per day}$

Emissions per year: $15 \times 223 = 3345 \text{ kg of CO}_2 \text{ per year}$

Annual emissions per person: $3345 \div 40 = 83.6 \text{ kg of CO}_2 \text{ per person (to 3 s.f.)}$



Task C: Calculating the cost of a journey

- 1) A person lives 6 miles from their place of work and travels by car 225 days per year.
On average, their car travels 45 miles per gallon.
At their nearest petrol station, petrol is sold for 148.9 p per litre.
1 gallon $\approx$ 4.546 litres.

- a) How much do they spend on petrol per year for **commuting** by car?

Example solution strategy:

$$\text{Miles per year: } 6 \times 2 \times 225 = 2700 \text{ miles per year}$$

$$\text{Gallons per mile: } 1 \div 45 = 0.02 \text{ gallons per mile}$$

$$\text{Gallons per year: } 0.02 \times 2700 = 60 \text{ gallons per year}$$

$$\text{Litres per year: } 60 \times 4.546 = 272.76 \text{ litres per year}$$

$$\text{Cost per year: } 272.76 \times 1.489 = \pounds406.14 \text{ per year}$$

- b) Their car insurance costs £65 per month.
Their annual maintenance costs sum to £820.
Assume that they only use the car for **commuting**.
How much do they spend in total on their car each year?

$$406.14 + 12 \times 65 + 820 = \pounds2006.14$$

- c) They are considering selling their car and **commuting** by bus.
A weekly bus pass in their town costs £22 per week.
Assume that they would buy a weekly bus pass every week of the year.
Would **commuting** by bus cost more or less money per year? Justify your answer.

$$\text{Annual cost of buying weekly bus passes: } 22 \times 52 = \pounds1144$$

It would cost less per year.

